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VESSEL WITH COMPOSITE HANDLE

Abstract

A vessel includes a container, a bracket, and a brace. The container has a cavity. The bracket extends from the container and includes an arm spaced from the container. The arm has opposed first and second ends, and the arm is connected to the container only at its first end. The brace connects the first end of the arm and the container. A first plurality of structural reinforcement flanges extend along the container between the brace and the second end of the arm. The first plurality of structural reinforcement flanges have curved exterior surfaces.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims the benefit of priority of U.S. Provisional Patent Application No. 63/313,456, filed Feb. 24, 2022; the priority application is hereby incorporated by reference in its entirety.

BACKGROUND

[0002] Vessels, containers, or trays are utilized for carrying a variety of materials or fluids. Typically, a handle is provided, which allows a user to carry or hold the container without contacting the fluid therein. A relative large container with a sturdy handle configuration is useful in many commercial or household uses and is especially useful in painting applications.

SUMMARY

[0003] In one aspect, a vessel comprises a container, a bracket, a bail and a grip. The container has a cavity configured to hold a liquid. The bracket extends from the container and comprises an arm spaced from the container. The arm has opposed first and second ends, and the arm is connected to the container only at its first end. The bail is pivotally connected to the container. The grip is disposed on at least a portion of the bail. The vessel is configured so that the grip has a position that is in contact with the arm, thereby defining, in combination, a composite vessel handle.

[0004] In another aspect, a method of supporting a vessel is described. The vessel comprises a container, a bracket, a bail and a grip. The container has a cavity configured to hold a liquid. The bracket extends from the container and comprises an arm spaced from the container. The arm has opposed first and second ends, and the arm is connected to the container only at its first end. The bail is pivotally connected to the container. The grip is disposed on at least a portion of the bail. The method comprises pivoting the bail so that the grip is in contact with the arm and positioning at least a portion of a hand on the grip and the arm.

[0005] This summary is provided to introduce concepts in simplified form that are further described below in the Detailed Description. This summary is not intended to identify key features or essential features of the disclosed or claimed subject matter and is not intended to describe each disclosed embodiment or every implementation of the disclosed or claimed subject matter. Specifically, features disclosed herein with respect to one embodiment may be equally applicable to another. Further, this summary is not intended to be used as an aid in determining the scope of the claimed subject matter. Many other novel advantages, features, and relationships will become apparent as this description proceeds. The figures and the description that follow more particularly exemplify illustrative embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] The disclosed subject matter will be further explained with reference to the attached figures, wherein like structure or system elements are referred to by like reference numerals throughout the several views. All descriptions are applicable to like and analogous structures throughout the several embodiments, unless otherwise specified.

[0007] FIG. **1** is a perspective view of an exemplary embodiment of a vessel, from the top, front and left side thereof.

[0008] FIG. **2** is a perspective view thereof, from the bottom, rear and right side.

[0009] FIG. **3** is a perspective view thereof, from the top, rear and left side.

[0010] FIG. **4** is a side view thereof (as taken from the right side in FIG. **1**).
[0011] FIG. **5** is a side view thereof (as taken from the left side in FIG. **1**), with a bail thereon pivoted to the front of the vessel.
[0012] FIG. **6** is a rear view of the vessel of FIG. **1**.
[0013] FIG. **7** is a front view thereof.
[0014] FIG. **8**A is a top view thereof.
[0015] FIG. **8**B is a cross-sectional view of FIG. **8**A taken along line B-B.
[0016] FIG. **9** is a bottom view of the vessel of FIG. **1**.
[0017] FIG. **10** is similar to FIG. **1** but shows a person holding the vessel by a composite handle and a paint roller held in a cavity of the vessel.
[0018] FIG. **11** is similar to FIG. **2** and shows a person holding the vessel by inserting fingertips into an underhook of the vessel and grasping the rear wall with a palm and thumb, with the front of the hand resting against at least the bail of the composite handle.
[0019] FIG. **12** is an enlarged perspective view of a ladder bracket of the vessel.
[0020] FIG. **13** is a perspective view of the vessel supported on a ladder rung and side rail by its ladder bracket.
[0021] FIG. **14** is a perspective view of a liner configured to fit into a cavity of the vessel.
[0022] While the above-identified figures set forth one or more embodiments of the disclosed subject matter, other embodiments are also contemplated, as noted in the disclosure. In all cases, this disclosure presents the disclosed subject matter by way of representation and not limitation. It should be understood that numerous other modifications and embodiments can be devised by those skilled in the art that fall within the scope of the principles of this disclosure.
[0023] The figures may not be drawn to scale. In particular, some features may be enlarged relative to other features for clarity. Moreover, where terms such as above, below, over, under, top, bottom, side, right, left, vertical, horizontal, etc., are used, it is to be understood that they are used only for ease of understanding the description. It is contemplated that structures may be oriented otherwise.
[0024] The terminology used herein is for the purpose of describing embodiments, and the terminology is not intended to be limiting. Unless indicated otherwise, ordinal numbers (e.g., first, second, third, etc.) are used to distinguish or identify different elements or steps in a group of elements or steps and do not supply a serial or numerical limitation on the elements or steps of the embodiments thereof. For example, "first," "second," and "third" elements or steps need not necessarily appear in that order, and the embodiments thereof need not necessarily be limited to three elements or steps. Unless indicated otherwise, any labels such as "left," "right," "front," "back," "top," "bottom," "forward," "reverse," "clockwise," "counter clockwise," "up," "down," or other similar terms such as "upper," "lower," "aft," "fore," "vertical," "horizontal," "proximal," "distal," "intermediate" and the like are used for convenience and are not intended to imply, for example, any particular fixed location, orientation, or direction. Instead, such labels are used to reflect, for example, relative location, orientation, or directions. The singular forms of "a," "an," and "the" include plural references unless the context clearly dictates otherwise.

DETAILED DESCRIPTION

[0025] This disclosure relates to a holding vessel and more particularly to a container configured for use with a paint roller having a length of about nine inches. The vessel includes a ladder bracket and a bail, wherein a bail grip can be positioned relative to the ladder bracket to form a composite handle.
[0026] FIGS. **1-11** show various views of an exemplary embodiment of a vessel **20** having a container **21**, bracket **22** and bail **24**. In some cases, this disclosure refers to a ladder bracket or bail handle. However, these descriptions of suitable uses are not intended to limit the structures to the uses described. It is contemplated that the vessel can be supported by structures other than a ladder or a user's hand, such as a scaffold bar, ground surface, joist, or other support, for example.
[0027] In an exemplary embodiment, container **21** includes a bottom wall **26** connected to front

wall **44**, rear wall **46** and side walls **28**. In an exemplary embodiment, bail **24** is manufactured separately from container **21** and is attachable thereto. In an exemplary embodiment, ladder bracket **22** is formed integrally with container **21**. In an exemplary embodiment, vessel **20** is made of polypropylene (or other suitable material, preferably plastic) to withstand the harmful effects of paint, stain or varnish. Recycled plastics, such as from car battery cases, may be used. The material in an exemplary embodiment is non-corrosive to prevent the formation of rust from repeated use and cleaning. Injection molding is a suitable manufacturing process for the formation of vessel **20**. Although the discussion herein will focus on paint, it is understood that the described vessel **20** can also be used to hold other fluid materials, such as another surface coating such as a protectant or varnish; an adhesive; a roofing, pavement or driveway coating; or a textured surface material, for example.

[0028] In an exemplary embodiment, bail **24** is formed of a metallic wire and has a central grip **30** made of an elastomer, rubber, or other flexible, yet durable material. In an exemplary embodiment, bail **24** attaches to a rim **60** of container **21** by insertion of ends **32** into apertures **34**. In an exemplary embodiment, bail **24** has bends **25** on either side of grip **30** to retain grip **30** in a central location on bail **24**, to overlay bracket **22** when pivoted toward a back of container **21**. In an exemplary embodiment, rim **60** extends laterally and downwardly from walls **28**, **44**, **46** to form a lightweight, structurally strengthening top perimeter feature.

[0029] In an exemplary embodiment, container **21** has a continuous inner surface **36** that includes inner surfaces of bottom wall **26**, front wall **44**, rear wall **46** and side walls **28**; the inner surface **36** defines a cavity of vessel **20** for carrying, holding or transporting loose materials or fluids, such as paint, stain or varnish. In an exemplary embodiment, vessel **20** includes a retainer for keeping a paintbrush or other tool within container **21**, preferably raised from the floor **26** of the cavity. In an embodiment, the retainer is at least one magnet **42**, which can be exposed in the cavity of container **21**. In another embodiment, magnet **42** is not necessarily exposed, but its magnetic force is active in the cavity of container **21**, such as through a thin layer of material. A user can position a metallic ferrule portion of a tool such as paintbrush (not shown) against or proximate magnet **42**, such as with the bristles of the paintbrush disposed inside the cavity of container **21**, such that paint on the bristles drips into the cavity.

[0030] A user can insert a disposable liner **38** in the cavity of vessel **20**. As shown in FIG. **14**, an exemplary liner **38** is thin but relatively rigid, so that it does not deform in use, as a plastic film bag might. An exemplary liner **38** has contours that closely match those of inner surface **36** of vessel **20**, so that use of liner **38** does not adversely affect the effective capacity of vessel **20**. Moreover, a material of liner **38** suitably does not adversely affect the ability of magnet **42** to attract and retain a metallic tool portion, even when the liner **38** is interposed between the magnet **42** and the tool. In an exemplary embodiment, liner **38** is formed of a lightweight plastic material and can be used as a disposable or semi disposable component. Liner **38** has contours that fit closely to the inner surface **36** of vessel **20**, so that its use does not interfere with paint distribution features of the vessel **20**.

[0031] In an exemplary embodiment, each of front wall **44**, rear wall **46** and side walls **28** inclines outwardly from bottom wall **26**, so that an upper portion of the cavity of vessel **20** has a larger lateral cross-section dimension than a lower portion. The larger cross-sectional dimension of the upper portion allows for easy access into the cavity of vessel **20** for insertion of a tool such as a paint brush or paint roller **40** (shown in FIG. **10**). The relatively smaller cross-sectional dimension of lower portion increases a depth of liquid at the bottom of container **21** for easy access by the tool. Moreover, the overall angled orientations of the walls **28**, **44**, **46** allow for nesting of multiple vessels **20**, thereby providing for space savings in storage, transport and merchandise display.

[0032] In an exemplary embodiment, each of front wall **44** and rear wall **46** provides a textured rolling surface for a paint brush or paint roller **40** (shown in FIG. **10**). In an exemplary embodiment, front wall **44** comprises a plurality of ridges **48**, each ridge **48** configured as a relatively pronounced curvilinear element. In an exemplary embodiment, rear wall **46** comprises a plurality of ridges **50**, each ridge **50** configured as a relatively shallow rectilinear element. Thus, the front and rear walls **44**, **46** offer differently textured surfaces against which a user may roll or brush a tool to evenly distribute paint thereon and remove excess paint therefrom.

[0033] Paint roller **40** can include roller **52**, frame **54** and handle **56**. Typically, a user dips paint roller **40** into a pool of paint or other fluid contained within vessel **20** to coat roller **52** with the paint or fluid. The user can then press roller **52** against front or rear wall **44**, **46** while moving handle **56** up and down to remove excess paint from roller **52** and more evenly distribute the paint upon the surface of roller **52**. Another tool, such as a paint brush (not shown) can similarly be moved with its bristles against ridges **48**, **50** to remove excess paint from the brush and more evenly distribute the paint upon the surface of its bristles. The excess paint is captured by ridges **48**, **50** and flows by gravity back into the pool of fluid within container **21**. Thus, a user can apply a uniform load of paint on a paint tool.

[0034] As shown in FIGS. **1** and **2**, in an exemplary embodiment, bottom wall **26** is substantially flat and rectangular, offering a large, stable surface on which vessel **20** may rest on a horizontal surface such as a floor or table, for example. Another manner in which vessel **20** can be supported is by bail **24**, which is pivotally connected to rim **60** at apertures **34**. As with a pail, bail **24** can swing from a rear of vessel **20** to a front side thereof. Most of the drawing figures show bail **24** pivoted so its grip **30** rests on ladder bracket **22**. However, as shown in FIGS. **5** and **13**, bail **24** is be pivoted to a forward position, wherein the bail **24** clears the front of rim **60** so that grip **30** rests against front wall **44**.

[0035] In an exemplary embodiment, grip **30** has a substantially triangular cross-sectional shape having three substantially flat surfaces **58**. In an exemplary embodiment, bail **24** passes through bore **86** of grip **30**, which freely spins about the inserted bail portion. Thus, when bail **24** is pivoted to a rear of vessel **20** so that grip **30** overlies ladder bracket **22**, one of the flat surfaces **58** rests stably against the ladder bracket **22**. Moreover, the spinning grip **30** of bail **24** allows container **21** to hang freely from bail **24**, no matter what grasping orientation a user places on grip **30**. In a configuration as shown in FIGS. **1-4** and **6-11**, the ladder bracket **22** and bail grip **30** together form a composite handle by which a user can firmly support vessel **20**, as shown in FIG. **10**, with a hand position that surrounds both the grip **30** and portions of the ladder bracket **22**. Another user engagement position on the composite handle is shown in FIG. **11**, wherein the user's palm and thumb grip rear wall **46** as the front of the hand contacts at least the grip **30** of the composite handle, while his or her fingertips curl into underhang **62**. At underhang **62**, rear wall **46** extends inward into an interior of container **21** to form a depression on the exterior or rear wall **46** into which a user may insert one or more fingertips.

[0036] FIG. **12** is an enlarged perspective view of ladder bracket **22**. In an exemplary embodiment, ladder bracket **22** is configured to extend from rear wall **46** and rim **60** and is connected to container **21** by an upwardly inclined brace **64**. Arm **66** extends from brace **64** and has a longitudinal extent that is aligned substantially parallel to an orientation of rim **60** at rear wall **46**. Arm **66** terminates at end protrusion **68**, which is opposite brace **64**. In an exemplary embodiment, arm **66** has a generally triangular cross-sectional shape, with two inclined front surfaces **70** and a back surface **71**. Each of the two inclined front surfaces **70** is covered with a plurality of ridges **72**.

[0037] FIG. **13** is a perspective view showing vessel **20** supported by ladder bracket **22** on a ladder **74** having side rails **76** and one or more rungs **78**. In FIG. **13**, bail **24** is swung forward as in FIG. **5**. A bottom surface of arm **66** rests on a top surface of rung **78** so that arm **66** is on one side of rail **76** and container **21** is on an opposite side of the same rail **76**. The ridges **72** act as surface texturing elements to grip and prevent sliding along. rung **78**. Arm **66** is sized to extend completely around a side rail **76**. Thus, when vessel **20** is moved so that brace **64** contacts the front surface of rail **76**, protrusion **68** extends beyond a back surface of rail **76**. Vessel **20** could be attached to the left rail **76**, in which case the brace **64** would be proximate the back surface of a rail **76** as illustrated. End protrusion **68** acts as a hook to prevent unintentional disengagement of the ladder bracket **22** from side rail **76**. The inclination of brace **64** allows container **21** to hang from ladder bracket **22** in a relatively level manner while accommodating various thicknesses of side rail **76**. Arm **66** and grip **30** are also sized to complement each other, so that one of surfaces **58** of grip **30** rests stably along the ridges **72** of the top front inclined surface **70**, between brace **64** and end protrusion **68**. As shown in FIG. **2**, reinforcing flanges **80** are provided on rear wall **46** connecting to rim **60** to reinforce and provide additional strength on a portion of the vessel **20** that rests against side rail **76**.

[0038] In an exemplary embodiment, the front of rim **60** includes protrusions **82** configured to serve as retaining stops for tools held in container **21**. For example, as shown in FIG. **10**, paint roller **40** is inserted into container **21**. To hold the paint roller **40** in place while the user moves vessel **20** by its composite handle, frame **54** of the paint roller **40** is laid against protrusion **82** to maintain a desired orientation of paint roller **40** within the large cavity of vessel **20**. Thus held, frame **54** is likely to remain in position between protrusion **82** and the closest side wall **28**, even as vessel **20** is maneuvered. As shown in FIG. **8**A, in an exemplary embodiment, pouring spouts **84** are integrally formed into rim **60** at a corner between front wall **44** and a respective side wall **28**.

[0039] The described vessel **20** can be supported in numerous ways. For example, it can be placed with its bottom wall **26** on a horizontal surface such as a floor or counter, for example. Vessel **20** can be carried as a pail, by a user's fingers curled about grip **30**, so that the vessel is suspended by its bail **24**. As shown in FIG. **13**, it can be hung by ladder bracket **22** from a side rail **76** of a ladder **74**, supported on rung **78**. And as shown in FIGS. **10** and **11**, it can be supported by a user at a composite handle formed of an overlaid grip **30** of bail **24** on arm **66** of ladder bracket **22**. While certain methods of use are hereby described, it is contemplated that users may find other ways to carry or move vessel **20**, such as suspending it from structures other than a ladder, for example.

[0040] Although the subject of this disclosure has been described with reference to several embodiments, workers skilled in the art will recognize that changes may be made in form and detail without departing from the scope of the disclosure. In addition, any feature disclosed with respect to one embodiment may be incorporated in another embodiment, and vice-versa. All references mentioned in this disclosure are hereby incorporated by reference.

Claims

1. A vessel comprising: a container having a cavity a bracket extending from the container, the bracket comprising an arm spaced from the container, the arm having opposed first and second ends, wherein the arm is connected to the container only at its first end; and a brace connecting the first end of the arm and e container; wherein a first plurality of structural reinforcement flanges extend along the container between the brace and the second end of the arm, and wherein the first plurality of structural reinforcement flanges have curved exterior surfaces.

2. The vessel of claim 1, wherein the brace extends upwardly from the container.

3. The vessel of claim 1, wherein the brace extends outwardly from the container.

4. The vessel of claim 3 comprising: a bail pivotally connected to the container; and a grip disposed on at least a portion of the bail; wherein the second end of the arm comprises a protrusion.

5. The vessel of claim 4 wherein: the arm has a first length between the brace and the protrusion; and the grip has a second length that is substantially equal to or less than the first length.

6. The vessel of claim **44** wherein the grip is disposed on a central portion of the bail.

7. The vessel of claim 4, wherein the grip has a substantially triangular cross-sectional shape.

8. The vessel of claim 1, wherein the arm has a substantially triangular cross-sectional shape.

9. The vessel of claim 1, wherein a portion of the container proximate the brace comprises a second plurality of structural reinforcement flanges.

10. (canceled)

11. The vessel of claim 1, wherein the container comprises a bottom wall connected to a front wall, a back wall, and two side walls, each of the front wall, back wall and two side walls extending upwardly and outwardly from the bottom wall so that a cross-sectional dimension of a top rim of the container is greater than a cross-sectional dimension of a bottom portion of the container.

12. The vessel of claim 11, wherein at least one of the front wall, back wall, and side walls comprises a plurality of ridges in an interior surface.

13. The vessel of claim 11 wherein the bracket extends from the back wall of the container, and wherein the back wall comprises an underhang that is spaced from the top rim and that is configured as an inward depression on an exterior side of the back wall.

14. The vessel of claim 11 wherein the bracket extends from the back wall of the container, and wherein the back wall comprises a magnet.

15. The vessel of claim 1 comprising a liner disposed in the container.

16.-**20**. (canceled)

21. The vessel of claim 1 comprising a bail pivotally connected to the container.

22. The vessel of claim 21 comprising a grip disposed on at least a portion of the bail.

23. The vessel of claim 22, wherein the grip is disposed on a central portion of the bail.

24. The vessel of claim 1 comprising a plurality of ridges extending from the arm.

25. The vessel of claim 24, wherein each of the plurality of ridges has a curved exterior surface.

26. The vessel of claim 24, wherein the plurality of ridges extend parallel to the first plurality of structural reinforcement flanges.